# Mohamed Nadhir Djekidel

**Ph.D. / Staff Scientist**

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## Professional Summary

Bioinformatics scientist specializing in multi-omics integration, with over 8 years of experience developing algorithms and automated workflows for transcriptomics, epigenomics, genomics, single-cell and spatial transcriptomics. Focused on transforming complex biological datasets into meaningful discoveries through following the best practices in the field. Successfully published in high profile journals such as Nature, Nature Genetics and Nature Neuroscience.

## Education

10/2012 - 04/2016 | **Tsinghua University**, Beijing, China
*Ph.D. in Bioinformatics (automation department)*

10/2009 - 07/2012 | **Central South University**, Changsha, China
*M.Sc, Computer Science (Bioinformatics)*

10/2008 - 07/2009 | **Changsha, China**, Changsha, China
*Certificate, Chinese (Mandarin) language*

10/2003 - 06/2008 | **Ecole Nationale Supérieure d'Informatique (ex I.N.I)**, Algiers, Algeria
*Eng, Computer Science (Computing Systems)*

## Professional Experience

08/2023 - Present | **Sidra Medicine (Perinatal and Reproductive Health Division)**
*Staff Scientist - Assistant level*
Lead the bioinformatics analysis, development and validation of different bioinformatics pipelines for the analysis of different types of omics data including: Transcriptome, microbiome, epigenomic and genetics data. Get involved in grants applications, research papers and several academic activities. Training and supervision of junior staff members.

09/2020 - 07/2023 | **St. Jude Children's Research Hospital (Center of Applied Bioinformatics, Epigenetics group)**
*Sr. Bioinformatics Research Scientist*
Develop standard epigenetics data analysis and reporting pipelines including CUT&RUN, ATAC-seq, ChIP-seq, WGBS, Hi-C and single-cell data analysis pipelines (scRNA-seq, scATAC-seq, spatial transcriptomics). Develop ML methods for pediatric leukemia samples classification. Present in different educational seminars, scientific events and conferences. Collaborate with different PIs. Research topics: Neuroscience, Cancer and basic molecular science projects.

10/2016 - 09/2020 | **Harvard Medical School (Yi Zhang lab)**
*Postdoctoral research fellow*
Lead and co-lead different research projects at Dr. Yi Zhang's lab. Developed several specialized custom pipelines for the down-stream analysis of in-house low-input sequencing methods. Developed/applied several statistical and machine learning methods (e.g. HMM, ML, regression) to help in processing/interpretation of the data. Processing of different omics data (RNA-Seq, ChIP-Seq, CUT&RUN, ATAC-Seq, scRNA-Seq, WGBS, RRBS). Develop single-cell RNA-seq and spatial transcriptomics (MERFISH) analysis standard. Research topics: Preimplantation Embryo Development, Neuroscience (addiction).

## LLM Training & Evaluation Experience

1. **Expert solutions for hard bioinformatics problems** (Writer/Reviewer). Authored and reviewed human expert solutions to challenging bioinformatics tasks for LLMs.
2. **Black-box bioinformatics function design** (Top 50 contributor). Designed hidden bioinformatics functions that LLMs had to reverse-engineer.
3. **Synthetic-data bioinformatics challenges** (Top performer). Designed hard bioinformatics problems built on synthetic data, with detailed task descriptions and grading rubrics.
4. **Real-world-data bioinformatics challenges.** Designed hard bioinformatics problems built on real-world datasets, with detailed task descriptions and grading rubrics.
5. **Prompt-driven bioinformatics Docker images** (Writer/Reviewer). Designed and reviewed optimized bioinformatics Docker images built from natural-language prompts.

## Publications

1. Shailesh, H., **Djekidel MN**, Noor, S., Hayati, L., Worgall, S., Al Khodor, S., & Janahi, I. (2026). miRNA profiling shows shared signatures in pediatric asthma, obesity and their comorbidity. **Frontiers in Immunology**.
2. Tao, R., Erkek-Ozhan, S., Xu, B., Matsui, Y., Mittal, P., Smith, K.S., Li, Y., Filipovic, D., Xu, R., Gao, Q., et al. (incl. **Djekidel MN**) (2026). Divergent medulloblastoma chromatin states disclose KDM2B as a selective dependency. **Nature Genetics**.
3. Torrico, E.C., Kaptein, P., Laalouhmi, F., Mitsialis, V., El Guendouzi, S., El Khayari, A., El Baha, C., Htite, E.D., Chawluk, S., Gauthier, C.D., et al. (incl. **Djekidel MN**) (2026). High-fat diet causes rapid loss of intestinal group 3 innate lymphoid cells through microbiota-driven inflammation and mitochondrial stress. **Immunity**.
4. Garibova, S., Stickler, E., Al Ali, F., Abdulla, M.A., Sathappan, A., Da'as, S.I., Ghanem, L., **Djekidel MN**, Portman, R., & Avella, M. (2025). Cross-species insemination reveals mouse sperm ability to enter and cross the fish micropyle. **eLife**.
5. Ata, M., Choucair, F., **Djekidel MN**, Sousa Esteves, M., Al Ali, F., Nawaz, S., El Taha, L., Soloviov, O., Sathappan, A., Aliyev, E., et al. (2025). Mutational disruption of transcription factors binding and regulatory networks in a case of unexplained total fertilization failure. **Journal of Translational Medicine**.
6. Shailesh, H., Rao, R., Noor, S., Hayati, L., **Djekidel MN**, & Janahi, I. (2025). MicroRNA signatures of pediatric obesity-associated asthma: identifying molecular links between obesity and asthma in children. **Allergy** (abstract).
7. Al-Dasooqi, D.S., Shailesh, H., **Djekidel MN**, Alabduljabbar, S., Ahmed, S.H., Olabi, Y., Shallouf, N., Hani, Y., Janahi, I., & Terranegra, A. (2025). Dietary fiber and fatty acids may impact clinical outcomes in pediatric obesity-associated asthma: insights from the SOAP study. **Frontiers in Nutrition**.
8. Hyle, J., Qi, W., **Djekidel MN**, Rosikiewicz, W., Xu, B., & Li, C. (2025). Deciphering the role of RNA in regulating CTCF's DNA binding affinity in leukemia cells. **Genome Biology**.
9. Saadaoui, M., **Djekidel MN**, Murugesan, S., Kumar, M., Elhag, D., Singh, P., Kabeer, B.S.A., Marr, A.K., Kino, T., Brummaier, T., et al. (2025). Exploring the composition of the placental microbiome and its potential origin in preterm birth. **Frontiers in Cellular and Infection Microbiology**.

10. Norrie, J.L., Lupo, M.S., Little, D.R., Shirinifard, A., Mishra, A., Zhang, Q., Geiger, N., Putnam, D., **Djekidel MN**, Ramirez, C., et al. (2025). Latent epigenetic programs in Müller glia contribute to stress and disease response in the retina. **Developmental Cell**.

11. Yousif, G., Murugesan, S., **Djekidel MN**, Terranegra, A., Gentilcore, G., Grivel, J.C., & Al Khodor, S. (2025). Distinctive blood and salivary proteomics signatures in Qatari individuals at high risk for cardiovascular disease. **Scientific Reports**.

12. Tao, R., Erkek-Ozhan, S., Xu, B., Matsui, Y., Mittal, P., Smith, K., Li, Y., Gao, Q., Darrow, E., Xu, R., et al. (2025). EPCO-22. Divergent medulloblastoma chromatin states disclose KDM2B as a selective dependency. **Neuro-Oncology**.

13. Murugesan, S., Yousif, G., **Djekidel MN**, Gentilcore, G., Grivel, J.C., & Al Khodor, S. (2024). Microbial and proteomic signatures of type 2 diabetes in an Arab population. **Journal of Translational Medicine**.

14. Lim, B., Matsui, Y., Jung, S., **Djekidel MN**, Qi, W., Yuan, Z.F., Wang, X., Yang, X., Connolly, N., Pilehroud, A.S., et al. (2024). Phosphorylation of the DNA damage repair factor 53BP1 by ATM kinase controls neurodevelopmental programs in cortical brain organoids. **PLOS Biology**.

15. LaFlamme, C.W., Rastin, C., Sengupta, S., Pennington, H.E., Russ-Hall, S.J., Schneider, A.L., Bonkowski, E.S., Almanza Fuerte, E.P., Allan, T.J., Zalusky, M.P.G., **Djekidel MN**, et al. (2024). Diagnostic utility of DNA methylation analysis in genetically unsolved pediatric epilepsies and CHD2 episignature refinement. **Nature Communications**.

16. He, M., Zong, X., Xu, B., Qi, W., Huang, W., **Djekidel MN**, Zhang, Y., Pagala, V.R., Li, J., Hao, X., et al. (2024). Dynamic Foxp3–chromatin interaction controls tunable Treg cell function. **Journal of Experimental Medicine**.

17. Lan, X., Mi, T., Alli, S., Guy, C., **Djekidel MN**, Liu, X., Boi, S., Chowdhury, P., He, M., Zehn, D., et al. (2024). Antitumor progenitor exhausted CD8+ T cells are sustained by TCR engagement. **Nature Immunology**.

18. Tao, R., Erkek, S., Matsui, Y., Xu, B., Smith, K.S., Gao, Q., Li, Y., **Djekidel N**, Darrow, E., Kumar, R., et al. (2024). MDB-10. Divergent medulloblastoma chromatin states disclose KDM2B as a novel oncogenic dependency. **Neuro-Oncology**.

19. Bhattacherjee, A., Zhang, C., Watson, B.R., **Djekidel MN**, Moffitt, J.R., & Zhang, Y. (2023). Spatial transcriptomics reveals the distinct organization of mouse prefrontal cortex and neuronal subtypes regulating chronic pain. **Nature Neuroscience**.

20. Matsui, Y., **Djekidel MN**, Lindsay, K., Samir, P., Connolly, N., Wu, G., Yang, X., Fan, Y., Xu, B., & Peng, J.C. (2023). SNIP1 and PRC2 coordinate cell fates of neural progenitors during brain development. **Nature Communications**.

21. Hyle, J., **Djekidel MN**, Williams, J., Wright, S., Shao, Y., Xu, B., & Li, C. (2023). Auxin-inducible degron 2 system deciphers functions of CTCF domains in transcriptional regulation. **Genome Biology**.

22. Wright, S., Hu, J., Wang, H., Hyle, J., Zhang, Y., Du, G., Konopleva, M.Y., Kornblau, S.M., **Djekidel MN**, Rosikiewicz, W., et al. (2023). Interrogating bromodomain inhibitor resistance in KMT2A-rearranged leukemia through combinatorial CRISPR screens. **Proceedings of the National Academy of Sciences**.

23. Searcy, M.B., Larsen IV, R.K., Stevens, B.T., Zhang, Y., Jin, H., Drummond, C.J., Langdon, C.G., Gadek, K.E., Vuong, K., Reed, K.B., et al. (incl. **Djekidel MN**) (2023). PAX3-FOXO1 dictates myogenic reprogramming and rhabdomyosarcoma identity in endothelial progenitors. **Nature Communications**.

24. Shah, K., Ma, J., **Djekidel MN**, Song, G., Umeda, M., Fan, Y., Wu, G., & Klco, J. (2023). Gene Expression Machine Learning Models Classify Pediatric AML Subtypes with High Performance. **Blood**.

25. Chen, R., Liu, Y., **Djekidel MN**, Chen, W., Bhattacherjee, A., Chen, Z., Scolnick, E., & Zhang, Y. (2022). Cell type–specific mechanism of Setd1a heterozygosity in schizophrenia pathogenesis. **Science Advances**.

26. Melville, J., Lun, A., **Djekidel MN**, Hao, Y., Eddelbuettel, D., & van der Bijl, W. (2022). Package 'uwot'. **CRAN**.

27. Wang, Y., Wang, M., **Djekidel MN**, Chen, H., Liu, D., Alt, F.W., & Zhang, Y. (2021). eccDNAs are apoptotic products with high innate immunostimulatory activity. **Nature**.

28. Xu, B., Wang, H., Wright, S., Hyle, J., Zhang, Y., Shao, Y., Niu, M., Fan, Y., Rosikiewicz, W., **Djekidel MN**, et al. (2021). Acute depletion of CTCF rewires genome-wide chromatin accessibility. **Genome Biology**.

29. Chen, Z.+, **Djekidel MN**^, & Zhang, Y. (2021). Distinct dynamics and functions of H2AK119ub1 and H3K27me3 in mouse preimplantation embryos. **Nature Genetics**.

30. **Djekidel MN**, Rosikiewicz, W., Peng, J.C., Kanneganti, T.D., Hui, Y., Jin, H., Hedges, D., Schreiner, P., Fan, Y., Wu, G., et al. (2021). CovidExpress: an interactive portal for intuitive investigation on SARS-CoV-2 related transcriptomes. **bioRxiv**.

31. Chen, R., Blosser, T.R., **Djekidel MN**, Hao, J., Bhattacherjee, A., Chen, W., Tuesta, L.M., Zhuang, X., & Zhang, Y. (2021). Decoding molecular and cellular heterogeneity of mouse nucleus accumbens. **Nature Neuroscience**.

32. Fu, X., **Djekidel MN**, & Zhang, Y. (2020). A transcriptional roadmap for 2C-like–to–pluripotent state transition. **Science Advances**.

33. Bhattacherjee, A., **Djekidel MN**, Chen, R., Chen, W., Tuesta, L.M., & Zhang, Y. (2019). Cell type-specific transcriptional programs in mouse prefrontal cortex during adolescence and addiction. **Nature Communications**.

34. Tuesta, L.M., **Djekidel MN**, Chen, R., Lu, F., Wang, W., Sabatini, B.L., & Zhang, Y. (2019). In vivo nuclear capture and molecular profiling identifies Gmeb1 as a transcriptional regulator essential for dopamine neuron function. **Nature Communications**.

35. Fu, X., Wu, X., **Djekidel MN**, & Zhang, Y. (2019). Myc and Dnmt1 impede the pluripotent to totipotent state transition in embryonic stem cells. **Nature Cell Biology**.

36. Chen, W., Chen, R., **Djekidel MN**, Bhattacherjee, A., Tuesta, L.M., & Zhang, Y. (2019). Single-cell RNA-SEQ of mouse nucleus accumbens reveals a subtype of D1 medium spiny neurons. **Journal of Neurochemistry**.

37. **Djekidel MN**, Inoue, A., Matoba, S., Suzuki, T., Zhang, C., Lu, F., Jiang, L., & Zhang, Y. (2018). Reprogramming of chromatin accessibility in somatic cell nuclear transfer is DNA replication independent. **Cell Reports**.

38. **Djekidel MN**, Chen, Y., & Zhang, M.Q. (2018). FIND: Differential chromatin interactions detection using a spatial Poisson process. **Genome Research**.

39. Li, Y., He, Y., Liang, Z., Wang, Y., Chen, F., **Djekidel MN**, Li, G., Zhang, X., Xiang, S., Wang, Z., et al. (2018). Alterations of specific chromatin conformation affect ATRA-induced leukemia cell differentiation. **Cell Death & Disease**.

40. Liang, Z., Li, G., Wang, Z., **Djekidel MN**, Li, Y., Qian, M.P., Zhang, M.Q., & Chen, Y. (2017). BL-Hi-C is an efficient and sensitive approach for capturing structural and regulatory chromatin interactions. **Nature Communications**.

41. **Djekidel MN**, Wang, M., Zhang, M.Q., & Gao, J. (2017). HiC-3DViewer: A new tool to visualize Hi-C data in 3D space. **Quantitative Biology**.

42. He, C., Li, G., Nadhir, D.M., Chen, Y., Wang, X., & Zhang, M.Q. (2016). Advances in computational ChIA-PET data analysis. **Quantitative Biology**.

43. Gao, J., Yang, X., Nadhir Djekidel, M., Wang, Y., Xi, P., & Zhang, M.Q. (2016). Developing bioimaging and quantitative methods to study 3D genome. **Quantitative Biology**.

44. **Djekidel MN**, Liang, Z., Wang, Q., Hu, Z., Li, G., Chen, Y., & Zhang, M.Q. (2015). 3CPET: finding co-factor complexes from ChIA-PET data using a hierarchical Dirichlet process. **Genome Biology**.

45. Khalid, H.G., Zuping, Z., Soliman, M.S., & **Djekidel MN** (2011). The design and implementation of integrated E-Cart on E-Hyper Market. **Global Journal of Computer Science and Technology**.

## Honors & Awards

2024 | **Best Poster Award**, MENA Microbiome Conference

2022 | **2nd Place, St. Jude BioHackathon**, St. Jude Children's Research Hospital

2019 | **Scientific Retreat Best Poster Award**, Harvard Medical School

2013 - 2016 | **Chinese Scholarship Council Scholarship**, Chinese Scholarship Council

2015 | **Best Poster Award**, Genomics Frontiers Symposium

2012 - 2013 | **Tsinghua University Scholarship**, Tsinghua University

2018 | **Top 10 Chinese Bioinformatics Methods in 2018**, Proteomics and Bioinformatics Journal

## Patents

2018 | **Patent: Chromatin interaction difference analysis method and system**
CN201810069870.2

2015 | **Software Copyright: 3CPET – Prediction of chromatin interaction maintaining complexes**
2016SR110295

## Teaching & Seminars

2024 | **16S Data Analysis Workshop**, MENA Microbiome Conference
Workshop on 16S rRNA data analysis, Doha, Qatar.

2020 - 2022 | **Machine Learning Educational Seminars**, St. Jude Children's Research Hospital

2020 - 2023 | **Bioinformatics (Epigenetics & Single-Cell) Educational Seminars**, St. Jude Children's Research Hospital

## Technical Skills

**Programming and scripting languages:** R, Python, C++, Bash, CSS, AWK, LaTeX, JavaScript, HTML.

**Workflow management:** Bpipe, Nextflow, Docker, R targets pipeline.

**Machine learning and Data science:** Keras, scikit-learn, Caret, tidymodels.

**Genomics & Genetics:** WGS analysis, GATK, ONT long-read analysis, GWAS, rare-variant burden analysis, mobile element variant analysis.

**Transcriptomics:** bulk RNA-seq (STAR, Kallisto, RSEM), miRNA profiling, machine learning classification from expression data.

**Epigenomics:** ChIP-seq, CUT&RUN, CUT&TAG, ATAC-seq, WGBS, RRBS, Bismark, DNA methylation arrays (EPIC v2, ChAMP, limma, ComBat).

**3D Genome:** Hi-C, ChIA-PET, chromatin interaction methods (FIND, 3CPET, HiC-3DViewer).

**Single-cell & Spatial:** scRNA-seq, scATAC-seq, multiome, CellRanger, Seurat, scanpy, spatial transcriptomics (MERFISH).

**Microbiome & Metagenomics:** 16S rRNA amplicon analysis, shotgun metagenomics (bioBakery), viromics (ViroProfiler), phage-host interaction, resistome analysis, microeco, microbial networks (SPIEC-EASI, NetCoMi).

**Multi-omics Integration:** cross-omics network analysis integrating microbiome, transcriptome, methylation, metabolome, proteomics, and clinical data.

**Core Tools:** samtools, sambamba, bowtie, pysam, R/Bioconductor (Rsamtools, rtracklayer, GenomicRanges, GenomicFeatures).

**Cloud:** Google Cloud, AWS.

## Spoken Languages

Arabic (Native), French (Advanced), English (Advanced), Chinese (Intermediate).